Q2. Difference between A and AO* Algorithm*

Answer:

Feature	A* Algorithm	AO* Algorithm
Type	Best-first search	AND-OR graph search
Structure	OR graph	AND-OR graph
Goal	Find shortest path	Solve problem decomposition
Evaluation	f(n) = g(n) + h(n)	Uses cost + heuristic on AND/OR nodes
Use case	Pathfinding	Problem solving with subgoals

Example:

- **A***: Finding shortest route in maps
- **AO***: Solving problems like decision trees or planning

Q3. Water Jug Problem (3L and 4L → get 2L)

Answer:

Steps:

1. Fill 4L jug → (0,4)
2. Pour into 3L jug → (3,1)
3. Empty 3L → (0,1)
4. Transfer 1L → (1,0)
5. Fill 4L → (1,4)
6. Pour into 3L → (3,2)

✅ **Result: 2 litres in 4L jug**

State Representation:

State = (x, y)

- x = water in 3L jug
- y = water in 4L jug

Explanation:

This is a classic **state-space problem** solved using search techniques like BFS/DFS.

Q4. Heuristic Search & BFS vs DFS

Answer:

Heuristic Search:

Uses a function **h(n)** to estimate cost to goal.

Example:

Greedy Best First Search uses:

f(n) = h(n)

Used in:

- Pathfinding
- Games

Difference between BFS and DFS

Feature	BFS	DFS
Strategy	Level-wise	Depth-wise
Data structure	Queue	Stack
Completeness	Yes	Not always
Optimal	Yes (unweighted)	No
Space	High	Low

Q5. Hill Climbing Algorithm

Answer:

Definition:

A local search algorithm that moves toward better states (higher value).

Steps:

1. Start with initial state
2. Evaluate neighbors
3. Move to best neighbor
4. Repeat until no improvement

Drawbacks:

- Gets stuck in **local maxima**
- **Plateau problem** (flat region)
- **Ridge problem**

How to Overcome:

- Random restart hill climbing
- Simulated annealing
- Use memory (tabu search)